

## Dataset

| Property                                  | Detail                                |
|-------------------------------------------|---------------------------------------|
| Total Videos                              | 618 videos                            |
| Classes                                   | 8 Sinhala words                       |
| Split                                     | 70% Train / 10% Validation / 20% Test |
| After augmentation one class for the data | 215 video                             |

## System Architecture

The full pipeline has 4 stages:

RAW VIDEO → FRAME EXTRACTION → ENHANCEMENT → MODEL TRAINING → EVALUATION  
(.mp4) (32 frames/video) (per experiment) (MobileNetV3+LSTM) (Test Set)

### Model: MobileNetV3Large + LSTM

#### Two-Phase Training Strategy:

| Phase               | What Happens                                           | Why                                                            |
|---------------------|--------------------------------------------------------|----------------------------------------------------------------|
| Phase 1 (15 epochs) | Backbone is FROZEN. Only LSTM + Dense head trains.     | Warm up the classifier without destroying pre-trained features |
| Phase 2 (50 epochs) | Entire model UNFROZEN. Fine-tunes all layers together. | Adapt MobileNet features to Sinhala sign language domain       |

## The 4 Experiments

All experiments use the **same model, same dataset, same splits** — only the image enhancement pipeline changes.

### EXP1 — Baseline (No Enhancement)

- Raw videos → extract frames → train directly
- No YOLO cropping, no filtering, no colour correction
- Mixup augmentation only during training

### EXP2 — CLAHE + Gamma Correction

- **CLAHE** (Contrast Limited Adaptive Histogram Equalization): fixes dark/uneven lighting by boosting local contrast
- **Gamma Correction ( $\gamma=1.2$ )**: brightens the overall frame
- Both applied per-frame before saving

### EXP3 — Edge-Preserving Sharpening (Bilateral + Unsharp Mask)

- **Bilateral Filter**: smooths noise while keeping hand edges sharp
- **Unsharp Mask**: enhances fine edge details of hand shape
- Applied per-frame before saving

### EXP4 — Hybrid (Bilateral + CLAHE + Unsharp)

- Combination of all techniques: Bilateral → CLAHE → Unsharp Mask
- Hypothesis: combining all methods should give best result

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## 5. Experimental Results

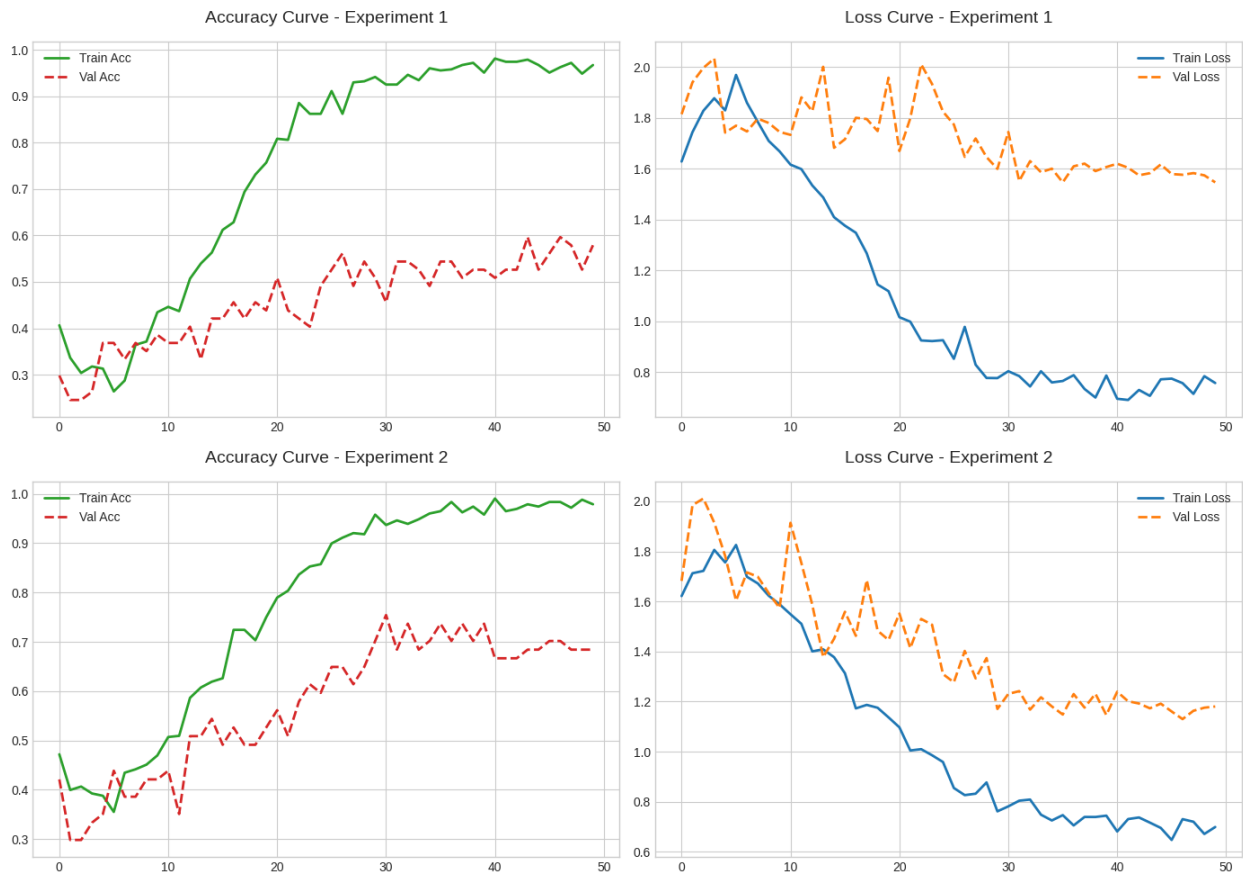
**Final Test Set Performance (133 unseen videos):**

| Experiment | Enhancement               | Top-1 Acc | Top-5 Acc | Macro F1 | Latency  |
|------------|---------------------------|-----------|-----------|----------|----------|
| EXP1       | Baseline (No Enhancement) | 54.14%    | 84.96%    | 0.5011   | 120.7 ms |
| EXP2       | CLAHE + Gamma ★           | 71.43%    | 95.49%    | 0.7047   | 139.3 ms |

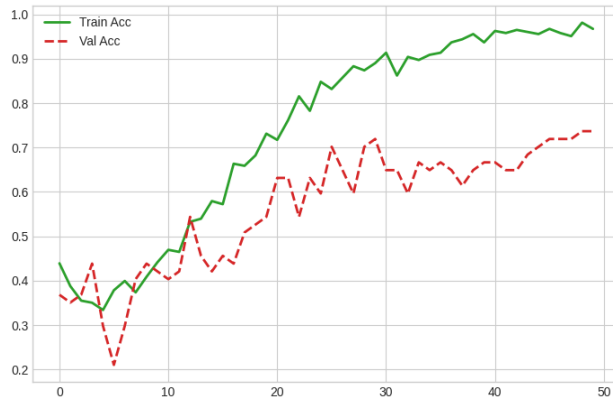
| Experiment | Enhancement           | Top-1 Acc | Top-5 Acc | Macro F1 | Latency  |
|------------|-----------------------|-----------|-----------|----------|----------|
| EXP3       | Bilateral + Unsharp   | 60.15%    | 89.47%    | 0.5716   | 137.3 ms |
| EXP4       | Hybrid (All combined) | 67.67%    | 91.73%    | 0.6506   | 172.4 ms |

## Key Findings:

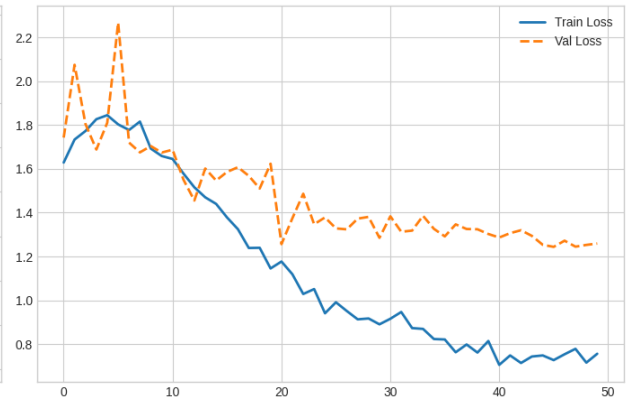
1. **✅ Image enhancement significantly improves accuracy** — EXP2 improved Top-1 accuracy by **+17.29 percentage points** over the Baseline.
2. **✅ CLAHE + Gamma (EXP2) is the best single technique** — simple, targeted contrast enhancement outperforms all others.
3. **✅ "More enhancement ≠ better accuracy"** — The Hybrid (EXP4) scored lower than CLAHE alone (EXP2). Over-processing creates visual artifacts that confuse the model.
4. **✅ All enhanced experiments beat the Baseline** — proving that image enhancement is a valid and necessary pre-processing step.



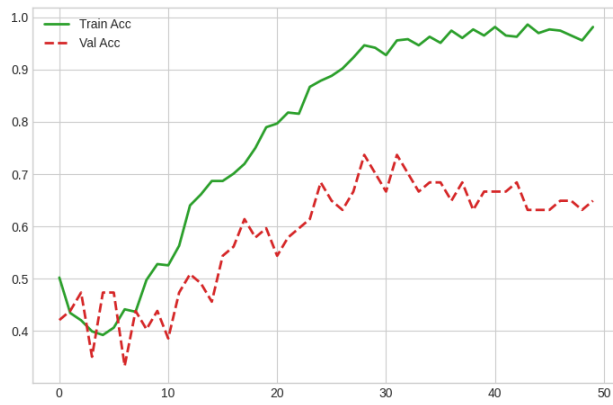
Accuracy Curve - Experiment 3



Loss Curve - Experiment 3



Accuracy Curve - Experiment 4



Loss Curve - Experiment 4

